A DEVELOPMENT OF AN AUTOMATED ENROLLMENT SYSTEM WITH PREDICTIVE ANALYTICS

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By

Baluyot, Danhil
Boado, Mark Frederick
Cardenas, Mharc Angelo
Casuncad, Westlie
Curtivo, Denmar

Eduardo S. Rodrigo
Capstone Adviser
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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

1.1 Project Context

The enrollment procedure is the most critical foundation of any educational institution because it connects student admission to active academic participation. It requires the careful handling of student records, required subjects, and schedule allocations to make sure school data is processed accurately. A well-organized enrollment setup is essential to maintain stability and guide the smooth flow of students into their correct programs before classes begin.

Traditionally, institutional enrollment systems heavily rely on decentralized, paper-based architectures that force students to physically route documents across multiple administrative nodes. This manual setup inevitably leads to severe workflow congestion, especially when cross-referencing legacy records, evaluating complex academic prerequisites, and clearing financial assessments. Furthermore, without an intelligent computational mechanism to forecast resource demands, academic institutions often struggle to optimize faculty loading and class section capacities. The reliance on manual estimation frequently results in misaligned academic schedules, dissolved classes, and poor tracking of student attrition, ultimately compromising the overall operational efficiency of the school.

To resolve these issues, this study proposes the development of an Automated Enrollment System with Predictive Analytics. This digital platform is designed to transform the paper-based workflow into a centralized and paperless system that automatically checks a student's past grades and required subjects. It features a digital portal where students can view their academic standing remotely, removing the need to present physical grade slips. Additionally, it uses predictive analytics to help the administration dynamically manage class capacities and forecast how many teachers will be needed in the future.

The main goal of this proposed system is to make the student registration experience faster while helping the school manage its resources better. It aims to eliminate repetitive manual steps, prevent errors like duplicate enrollments, and secure student records through a real-time tracking dashboard. Ultimately, the system strives to provide the school leaders with clear forecasting reports so they can track student dropouts, analyze retention rates, and allocate institutional resources efficiently.

1.2 Background of the Study

Global Reciprocal Colleges (GRC) is an educational institution in Caloocan City founded to provide affordable and quality education. Because GRC follows an "open admission policy," the student population is steadily growing. However, as the number of enrollees increases, the school's manual and paper-based administrative systems are struggling to scale with the demand.

Based on the official workflow diagrams provided by the institution, the existing setup forces old and regular students to execute a strict six-to-seven (6-7) step sequential workflow.

However, for new and transferee students, this process extends to an exhaustive eleven (11) step architecture due to the preliminary admission phase required before entering the core enrollment cycle. This overall process requires exhaustive physical routing across different floors and administrative nodes—such as navigating from the 3rd-floor Advising rooms down to the 2nd-floor Computer Laboratory for manual subject encoding, before proceeding to the Library, Registrar, Cashier, and Admission Office. Despite recent streamlining, this redundant physical routing and manual verification of Pre-Enrollment Forms (PEF) cause severe workflow congestion, resulting in measurable processing delays that span from several hours to multiple days per student before an official enrollment status is secured.

During structural interviews with key personnel, these operational inefficiencies were quantified to justify the need for system automation. The Registrar Head and the Academic Dean highlighted that evaluating irregular and returning students requires faculty members to manually cross-reference physical curriculum sheets and grade slips. Because the data is not centrally integrated, this manual verification loop heavily compromises data accuracy and severely limits the number of students the administration can process daily.

Finally, a critical systemic constraint lies in the failure of manual forecasting for institutional resource allocation. According to the IT Program Chair, estimating the required number of class sections during pre-enrollment relies heavily on unverified guesswork from previous years. This manual forecasting consistently fails when sections do not meet the strict 25-student minimum threshold, leading to the frequent dissolution of underfilled classes and misaligned faculty workloads. Therefore, the administration strictly requires an automated architecture integrated with predictive analytics to mathematically forecast section demands and replace the flawed manual estimation process.

1.3 Statement of the Problem

This study identifies that the existing manual enrollment system of Global Reciprocal Colleges (GRC) suffers from severe operational inefficiencies due to its decentralized and paper-based architecture. Specifically, this study seeks to address the following problems regarding the current implementation:

1. How does the current fragmented physical routing workflow, which spans six (6) to eleven (11) sequential steps, contribute to severe queuing delays and operational congestion across administrative floors?
2. To what extent does the manual cross-referencing of physical curriculum sheets and grade slips lead to evaluation errors and data inconsistencies during the academic advising of irregular and returning students?
3. How does the institutional reliance on unverified guesswork for forecasting section demands result in the frequent dissolution of underfilled class sections and the misalignment of faculty workloads?
4. In what way does the absence of a real-time data bridge between the Registrar and Accounting offices perpetuate the redundant manual verification of Pre-Enrollment Forms (PEF), thereby delaying the final financial clearance and payment phase?
5. What is the measured level of software quality of the proposed automated system—in terms of functional suitability, performance efficiency, and predictive accuracy—when evaluated against the ISO/IEC 25010 standard and standard error metrics?

1.4 Objectives of the Study

General Objective

The general objective of this study is to design and develop an Automated Enrollment System with Predictive Analytics for Global Reciprocal Colleges (GRC) that transforms the fragmented manual enrollment process into a centralized, data-driven digital platform to optimize administrative efficiency and resource allocation.

Specific Objectives

Specifically, the study aims to achieve the following deliverables to resolve the identified operational problems:

1. To consolidate the fragmented 6-to-11 step physical routing workflow into a centralized web-based enrollment portal, thereby eliminating the need for student navigation across multiple administrative floors and minimizing queuing delays.
2. To engineer an automated academic advising module capable of real-time prerequisite validation by cross-referencing digital academic histories against curriculum rules, effectively eliminating manual evaluation errors during the advising phase.
3. To build a predictive analytics and forecasting engine that utilizes historical enrollment data to mathematically project incoming section demands and optimize faculty loading, replacing the institutional reliance on manual guesswork.
4. To establish a real-time financial data bridge between the Registrar and Accounting modules to automate the verification of digital Pre-Enrollment Forms (PEF) and accelerate the financial clearance sequence.
5. To evaluate the system’s technical performance and functional suitability against the ISO/IEC 25010 software quality standard and verify the predictive engine’s accuracy using standard statistical error metrics.

1.5 Scope and Limitation

The scope of this study encompasses the architectural design, business logic mapping, and high-fidelity prototyping of the Automated Enrollment System with Predictive Analytics. To ensure strict alignment with the system architecture, the project's scope is explicitly bounded by the following functional modules, user roles, and artificial intelligence parameters:

1. User Roles and Module Access Boundaries The system's operational scope is strictly delimited to nine (9) predefined external entities, governed by a centralized Role-Based Access Control (RBAC) architecture. Their access is bounded to the specific core processes of the system:

- **Students and Admission Staff:** Bounded to *Process 2.0 (Process Enrollment and Digital Advising)*. Students can execute digital self-evaluation, receive automated subject recommendations, and generate a digital Queue Ticket. Admission Staff are delimited to view-only access and the initiation of New Student Profiles.
- **Program Chair, Professor, Dean, and Executive Director:** Bounded to *Process 1.0 (Establish Pre-Enrollment Schedules)*. Professors provide availability and encode grades, while the Program Chair manages curriculum capacities and faculty loading. The Dean and Executive Director are authorized to execute multi-level schedule approvals and monitor real-time dashboards.
- **Registrar Head and Registrar Staff:** Bounded to *Process 3.0 (Execute Final Approvals)*. The Registrar Head holds exclusive administrative clearance for override commands and final enrollment approvals, while the Registrar Staff handles transferee credit mappings and withdrawal forms.
- **Accounting Staff:** Bounded to the final verification phase, explicitly limited to view-only access of the pending payment dashboard and the management of the Active Serving Number.

Artificial Intelligence and Predictive Analytics Scope

The Artificial Intelligence (AI) feature of the system is strictly scoped to *Process 4.0 (Generate Predictive Analytics and Forecasting)*. The system utilizes supervised machine learning algorithms configured to ingest records from HISTORICAL DATA. The AI's capability is exclusively designed to:

- Forecast incoming class section demands based on historical enrollment trends.
- Provide empirical data for the Program Chair to optimize faculty load assignments.
- Generate predictive attrition analytics to help the Registrar Head identify at-risk students and dropout probabilities.

Delimitations of the Study

To prevent scope creep, the project excludes hardware-based integrations such as biometric attendance tracking, turnstile access, or Radio Frequency Identification (RFID) systems, as the focus remains strictly on software-based academic and enrollment management. Furthermore, the predictive analytics engine is explicitly delimited from utilizing Generative AI, Natural Language Processing (NLP) chatbots, and does not execute automated physical room timetabling or genetic algorithm-based scheduling.

Regarding financial transactions, while the system does not feature a direct online payment gateway (e.g., e-wallets or bank APIs) due to institutional transaction cost constraints, it directly resolves the payment queuing congestion by automating the financial assessment phase. The system automatically computes the required tuition fees and generates a digitally verified Queue Ticket that bridges real-time data to the Accounting Staff's view-only dashboard. This architectural design eliminates the need for cashiers to manually re-compute fees or re-verify physical Pre-Enrollment Forms (PEFs), thereby drastically accelerating the physical transaction throughput to formally release the student's Certificate of Registration (COR).

Finally, the actual software programming, full backend deployment, and live institutional database integration of the entire proposed system are specifically delimited to the future Capstone 2 development phase.

1.6 Significance of the Study

The findings and output of this study will be highly beneficial to the following stakeholders, arranged in alphabetical order:

Accounting Staff: The system directly resolves payment processing delays by providing a digitally verified Queue Ticket. By utilizing the system's real-time, view-only pending payment dashboard, the accounting personnel can efficiently process tuition fee payments and release the Certificate of Registration (COR) without manually re-computing the assessment.

Admission Staff: The system provides admission personnel with streamlined, view-only access to new student profiles. By digitizing the initial registration phase, it minimizes manual data entry while maintaining strict Role-Based Access Control (RBAC) to ensure they cannot inadvertently alter complex academic records.

Dean: The web-based platform delivers a real-time enrollment tracking dashboard that grants the Dean immediate visibility into the institution's overall academic progress, allowing for data-driven decisions during the preliminary approval of class schedules and faculty load assignments.

Executive Director: Acting as the highest level of the approval workflow, the system provides the Executive Director with the final authority to review, validate, and officially lock the master schedules and academic resource allocations before the enrollment period strictly begins.

Professors / Faculty: The system features a dedicated faculty grading module that allows professors to efficiently encode final grades digitally. This direct integration ensures that student academic records are instantly available for the system's automated prerequisite validation.

Program Chair: The system significantly alleviates the highly complex, manual burden of schedule preparation. By utilizing the predictive analytics engine, the Program Chair receives automated section demand forecasts, allowing for optimized faculty load assignments and the prevention of dissolved classes.

Registrar Head: The developed system provides the Registrar Head with full oversight and override controls to prevent double-enrollment anomalies. The integration of predictive attrition analytics and daily audit reports allows the Registrar Head to track dropouts, analyze retention rates, and maintain absolute operational control.

Registrar Staff: By automating the prerequisite validation and digital advising processes, the system relieves the registrar staff from manually cross-referencing physical grade slips against legacy curriculum mappings, accelerating the verification of clearances and transferee credits.

Student: As the primary end-users, the system completely eliminates the exhaustive 6-to-11 step physical routing for students. It provides a centralized web portal where they can execute digital self-evaluation, receive automated subject recommendations, and secure a digital Pre-Enrollment Form (PEF), drastically reducing physical queuing delays.

1.7 Definition of Terms

For a clearer understanding of this study, the following terms are operationally defined strictly based on how they function within the proposed System:

Automated Prerequisite Validation: An internal system sub-routine that automatically cross-references a student's academic history from the STUDENT RECORDS against the curriculum mapping rules to immediately evaluate eligibility without the manual checking of physical grade slips.

Digital Pre-Enrollment Form (PEF): The system-generated digital artifact that replaces the physical paper routing. It securely locks a student's validated subject selections and serves as the digital audit trail before the Registrar Head executes final enrollment approval.

Digital Queue Ticket: The active serving number automatically generated by the system after academic advising, which bridges the "Pre-Assessed Digital Handoff" by pushing view-only pending payment data directly to the Accounting Staff's dashboard.

Feedback Loop: The automated data routing mechanism where validated enrollment volumes and finalized faculty loads are continuously pushed back into HISTORICAL DATA at the end of each semester to train and recalibrate the predictive forecasting models.

Historical Enrollment Data: The specific collection of past transaction logs, student attrition records, and prior section capacities stored in HISTORICAL DATA, which acts as the mathematical baseline for the system's machine learning algorithms.

Predictive Analytics Engine: The specific analytical module of the system (Process 4) that ingests historical data to mathematically forecast future class section demands, prevent the dissolution of underfilled classes, and dynamically recommend schedules for irregular students.

Real-Time Enrollment Dashboard: The specific monitoring interface provided exclusively to the Dean, Executive Director, and Registrar Head that displays live queuing metrics, actual enrolled student counts, and active section capacities to support data-driven institutional decisions.

Role-Based Access Control (RBAC): The security architecture integrated into the system that strictly delimits data access among the nine (9) external entities, ensuring that the Admission Staff cannot alter academic grades and only the Registrar Head holds the clearance to execute enrollment override commands.

Rule-Based Advising: The logical framework used during the digital student self-evaluation phase that automatically restricts subject selection based on institutional constraints, specifically preventing irregular students from consuming slots in regular block sections.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Introduction

This chapter presents a review of related literature and studies that establish the conceptual, theoretical, and technical foundations for developing the Automated Enrollment System with Predictive Analytics. It explores key paradigms such as advanced data management, web-based architectures, the impact of system automation, and role-based access control. By synthesizing local and foreign materials, this chapter identifies the systemic gaps in traditional, manual enrollment methods and justifies the need for an intelligent, centralized, and data-driven platform to optimize the school's administration.

2.2 Review of Related Literature

Foreign Literature

According to Shafiq et al. (2022), Educational Data Mining (EDM) and learning analytics theories provide the technical foundation for extracting hidden patterns from historical academic records. Their research emphasizes that these mathematical methodologies enable the deployment of early warning algorithms to identify at-risk individuals before academic failure occurs. In accordance with the objectives of the proposed system, this theoretical research validates the use of predictive modeling within Process 4.0 as a fundamental tool for improving student management through data-driven interventions.

Based on the systematic review by de Oliveira et al. (2021), the effective application of learning analytics relies on the theoretical categorization of student features into sociodemographic and academic factors. The theory suggests that analyzing independent variables such as age, gender, previous academic history, and financial constraints provides a highly accurate algorithmic basis for understanding student attrition. This framework justifies the data extraction logic of the proposed system, ensuring that the predictive engine analyzes holistic variables from STUDENT RECORDS to calculate reliable dropout probabilities.

As stated by Abiola-Adams et al. (2025), strategic enrollment management is theoretically grounded in quantitative forecasting models that utilize historical institutional data. The application of statistical regression and decision tree algorithms allows institutions to identify consistent patterns and anomalies in matriculation rates and program-level distributions over multiple academic cycles. This theoretical perspective directly supports the forecasting engine of the proposed system, enabling the administration to mathematically estimate future section demands rather than relying on manual assumptions.

According to recent foreign literature on early dropout prediction of Park and Yoo (2021), analyzing the correlation between academic administration data and student learning behavior is essential for calculating re-enrollment probabilities. The theoretical approach of evaluating past grades against current academic standing allows educational administrators to determine whether a student is likely to persist or drop out in the succeeding semester. This concept provides the theoretical justification for the continuous updating of ENROLLMENT RECORDS in the proposed system to monitor student trajectories effectively.

In accordance with the digital academic advising framework discussed by Sundar et al. (2024) , the integration of rule-based constraints and knowledge grounding algorithms allows for the dynamic validation of curriculum policies. Theoretical approaches in intelligent student support dictate that mapping institutional regulations into machine-readable rules enables the automated verification of prerequisites and degree progress. This theoretical logic perfectly defends the core mechanism of Process 2.2 (Validate Capacities and Prerequisites) in the proposed system, ensuring that subject recommendations strictly adhere to the academic curriculum.

Based on the research by Khumalo et al. (2023), the theoretical concept of self-authorship in higher education is significantly enhanced when students are provided with technology-mediated advising platforms. Self-authorship theory posits that allowing students to independently track their academic trajectory fosters cognitive agency and self-regulation. This directly supports the user interface architecture of the proposed system, ensuring that students are empowered to execute digital self-evaluation seamlessly without absolute dependence on manual faculty advising.

As emphasized by Nnadozie et al. (2023), ecosystemic and attribution theories suggest that shifting from reactive interventions to proactive academic support significantly improves university outcomes. Understanding how students attribute meaning to their academic progress allows institutions to structure learning environments that leverage student strengths rather than merely addressing failures. This theoretical paradigm justifies the elimination of the fragmented 6-to-11 step physical routing in the proposed system, transforming enrollment into a proactive, student-centered digital experience.

According to the research of Yoheswari (2025) regarding Educational ERP

systems , Role-Based Access Control (RBAC) is a foundational cybersecurity framework that restricts resource access strictly according to organizational roles rather than individual identities. The theory of least privilege ensures that users, such as students or staff, are only granted the specific permissions necessary to execute their respective functions. This concept theoretically defends the security architecture of the proposed system's Context Diagram, ensuring that strict user boundaries are enforced to prevent unauthorized data manipulation.

Based on the comparative analysis by Mollakuqe and Dimitrova (2024), identity management theories assert that implementing granular access limitations and strict authentication protocols is mandatory to protect digital educational ecosystems. Their findings indicate that compartmentalizing privileges—such as assigning read-only access to specific staff while reserving administrative override capabilities for higher-level executives—mitigates the risk of inappropriate data exposure. This literature provides the precise structural baseline for the centralized RBAC implementation within the proposed system.

Furthermore, according to the research of **Singer (2023)** on Educational Data Mining (EDM), the integration of machine learning algorithms is highly effective in predicting online student enrollment decisions and academic outcomes. Their study demonstrated that predictive models evaluating independent student variables could accurately categorize enrollment trajectories with up to 81% accuracy. More importantly, the research concluded that enrollment managers strongly need these predictive frameworks to inform their strategic decision-making and resolve complex operational challenges. This literature provides a robust theoretical justification for the implementation of the proposed system's **Process 4.0 (Generate Predictive Analytics and Forecasting)**. By adopting a similar data-driven architecture, the proposed system guarantees that the Dean, Program Chair, and Registrar Head are equipped with an intelligent dashboard that mathematically forecasts section demands and student retention, effectively eliminating the institutional reliance on manual estimations.

Local Literature

According to Deleña et al. (2025), Educational Data Mining (EDM) provides the mathematical foundation for analyzing institutional data to predict student outcomes and inform strategic interventions. Their research establishes that integrating sociodemographic variables and academic records into machine learning models, specifically through the use of Extreme Gradient Boosting (XGBoost), yields superior accuracy in forecasting student retention. This theoretical approach justifies the core analytical engine of the proposed system, ensuring that attrition predictions are handled by robust algorithms capable of processing complex educational datasets.

Based on the research of Esquivel and Esquivel (2021), in estimating enrollment decisions, the theory of binary classification is a critical concept utilized by academic administrators to mitigate the uncertainty of human selection patterns. The utilization of Logistic Regression as a discriminative classifier allows institutions to calculate a predictive probability between 0 and 1, establishing a mathematical basis for determining whether an applicant will formally enroll. This concept theoretically supports the predictive forecasting capabilities of the proposed system, allowing the administration to proactively estimate incoming student populations.

As stated by Duran (2025), machine learning regression techniques provide an advanced methodology for modeling non-linear patterns in student academic and behavioral data. Unlike simpler linear models, these computational architectures adjust parameters to minimize prediction errors, making them highly reliable in differentiating between potential dropouts and retained students based on historical trends. This algorithmic framework theoretically defends the deployment of advanced predictive analytics within the proposed system.

According to Diaz and Jain (2025), the shift from intuition-based to evidence-based administrative operations in local universities relies heavily on the integration of predictive and prescriptive analytics. While predictive models forecast rising enrollments and potential attrition, prescriptive analytics logically recommends specific strategies for optimal resource allocation, such as faculty staffing and budget adjustments. The synergy of these analytical theories serves as the foundation for the proposed system's ability to not only project section demands but also prescribe actionable insights for the Program Chair and Dean.

In accordance with the findings of Grepon et al. (2021), Information Systems theory dictates that the centralization of data through a robust relational database framework is fundamental in eliminating manual paperwork and preventing data redundancy. The adoption of a client-server architecture ensures high functionality and reliability in data processing, enabling real-time transactions across different academic departments. This concept firmly supports the three-tier architecture of the proposed system, guaranteeing that records and enrollment transactions are securely processed and stored within a unified repository.

Based on the evaluation of Tan et al. (2024), resolving severe inefficiencies in inter-office operations requires Business Process Analysis and the use of logical modeling to streamline fragmented workflows. The mapping of academic transactions through use case architectures allows system designers to define strict user boundaries and eliminate redundant manual processes between the Registrar, students, and faculty. This theoretical modeling directly validates the design of the proposed system's Context Diagram, ensuring that every user interaction is systematically routed.

According to Berroa et al. (2025), the Technology Acceptance Model (TAM) serves as the primary theoretical framework for evaluating user acceptability in digital educational platforms. The construct of Perceived Usefulness (PU) dictates how students and administrators assess the efficiency and benefit of transitioning from manual methods to virtual enrollment processes. By integrating this framework, institutions can gauge whether a proposed academic portal successfully meets the operational needs of its users, providing a baseline for evaluating software quality.

As further emphasized in related local studies on system evaluation by **Tan et al. (2024)** , within the TAM framework, Perceived Ease of Use (PEOU) and behavioral intention are critical psychological factors that influence continuous user engagement. Local literature **Early Dropout Prediction in Online** highlights that if a digital system is overly complex or difficult to navigate, user satisfaction drops, leading to poor system adoption. This theoretical understanding requires that the presentation layer of the proposed system prioritize high-fidelity design and intuitive navigation, ensuring frictionless digital self-evaluation for the students.

As stated by Alimboyong and Bucjan (2021), the successful deployment of centralized academic portals is deeply connected to the theories of cloud computing infrastructure and institutional operational readiness. While cloud-based resources significantly improve operational scalability and data accessibility across multiple campuses, their adoption requires robust network bandwidth and strict identity management. These infrastructural concepts ensure that the real-time predictive analytics and enrollment transactions of the proposed system can be processed seamlessly without compromising data security.

Finally, according to Almonteros et al. (2022), the automation of complex academic constraints relies heavily on rule-based algorithms and backward chaining logic. Computational theories allow for the dynamic cross-referencing of a student's academic history against predefined curriculum rules, effectively eliminating the need for human intervention in detecting prerequisite violations. This algorithmic logic perfectly defends the automated digital advising module of the proposed system, ensuring that irregular students are provided with conflict-free subject recommendations.

2.3 Review of Related Studies

Foreign Studies

The development of predictive analytics in academic systems provides a strong architectural benchmark for forecasting enrollment behaviors. Ríos-Parra (2023) developed the **Course Enrollment Prediction Tool**, an interactive web-based decision support application built using the Dash framework in Python. The system evaluated multiple machine learning models and revealed that the Random Forest algorithm exhibited the best performance (AUC of 0.94) in automating semester-based course planning. This directly supports the architectural feasibility of the proposed system’s Process 4.0 (Generate Predictive Analytics and Forecasting). By mirroring the methodology of extracting past academic records to predict future section demands, the proposed system will similarly utilize HISTORICAL DATA to prevent class over-offering and under-offering, providing the Program Chair with empirical data for faculty loading.

To address the highly imbalanced profile of student attrition, **Prasad and Casale (2022) developed the Evasão PowerBI Dropout Prediction System**, an end-to-end machine learning platform. Using XGBoost and Random Forest algorithms, the system successfully processed student grades, tuition data, and socio-economic information to achieve AUC-PR scores of up to 89.5% in predicting dropouts across different trimesters. This study validates the deployment of supervised machine learning classifiers inside the proposed system's predictive engine, proving that mining structural data from STUDENT RECORDS and HISTORICAL DATA can accurately calculate dropout probabilities.

In a similar effort to proactively detect academic anomalies, Skittou et al. (2024) engineered a **Django-based Early Warning System (EWS)** with an MVC architecture. By utilizing the K-Nearest Neighbor (KNN) algorithm, the system achieved a 99.5% accuracy rate in processing socio-cultural and educational factors to predict at-risk students. This perfectly maps to the proposed system's Process 4.3 (Publish Forecasts and Dashboard). Just as the Django EWS utilized visual charts and geographic maps to alert administrators, the proposed system will supply the Registrar Head and Dean with real-time enrollment dashboards and stuck-student reports to coordinate timely academic interventions.

To resolve the workflow congestions inherent in manual curriculum evaluation, Jamaludin et al. (2025) designed the **UTMSPACE Academic Advising System**, a centralized web platform tailored for foundation programs. The implementation of the system successfully mitigated the limitations of traditional advising by providing role-specific dashboards equipped with structured checklist guidelines and real-time tracking of academic performance. This serves as the structural blueprint for the proposed system's Process 2.0 (Process Enrollment and Digital Advising), justifying the elimination of the fragmented 6-to-11 step physical routing in favor of a centralized portal where students can seamlessly execute digital self-evaluation.

Focusing on the transparency of artificial intelligence, **Prasad and Casale (2022)** produced the **Interactive Explainable AI Dashboard**. This specific module utilized SHapley Additive exPlanations (SHAP) values to identify and display the exact socio-economic and academic reasons behind a student's predicted dropout. This directly defends the Non-Functional Requirements of the proposed system regarding system interpretability. By integrating similar explainable methodologies, the proposed system ensures that the attrition analytics presented to the administration are not just arbitrary numbers, but actionable insights that pinpoint why a specific student is at risk.

Exploring the predictive power of admission data, Carballo-Mendívil et al. (2025) integrated an XGBoost-based early warning system into a university's **Student Trajectory Information System (SITE)**. The implemented system achieved an 88% sensitivity in detecting dropout risks from day one by exclusively mining pre-enrollment predictors such as high school GPA and socio-economic constraints. This study architecturally justifies the proposed system's Process 2.1 (Authenticate and Read Profile), proving that incoming student profiles collected by the Admission Staff can be immediately processed by the predictive engine to project enrollment risks even before classes officially start.

To automate complex academic constraints, Wagner et al. (2023) launched the **AIStudyBuddy** and **BuddyAnalytics** systems, which integrated process mining and rule-based Artificial Intelligence (Answer Set Programming). The system successfully modeled institutional examination regulations into logical rules to provide students with immediate, high-quality feedback regarding potential conflicts in their study plans. This study justifies the core logic of Process 2.2 (Validate Capacities and Prerequisites). Instead of faculty members manually cross-referencing legacy curriculum sheets, the proposed system will utilize an automated algorithm to dynamically cross-reference a student's academic history against CURRICULUM and SCHEDULING constraints, automatically generating conflict-free subject recommendations.

In optimizing university resources, Shao et al. (2021) implemented the **SDSU Course Enrollment Prediction Model** at San Diego State University. By applying Classification and Regression Trees (CART) and Random Forest algorithms to historical academic and demographic datasets, the system effectively predicted the enrollment volume for large-scale courses. This directly defends the data extraction logic of the proposed system’s forecasting module. Similar to the SDSU model, the proposed system will aggregate past enrollment volumes to estimate future section demands, providing the Program Chair with a mathematically sound baseline rather than relying on unverified guesswork.

Targeting administrative intervention, Guzmán-Castillo et al. (2022) deployed **SATD (Sistema de Alerta Temprana de Deserción)**, a predictive information system designed to calculate dropout risks using Random Forest algorithms. The system categorized students into risk quintiles, and its alert generation procedure successfully increased administrative intervention coverage by 15.2%. This validates the continuous feedback loop established in the proposed system architecture. By automatically pushing predictive attrition analytics to the Registrar Head, the system translates raw dropout forecasts into prioritized, data-driven institutional retention strategies.

Finally, emphasizing long-term trajectory analysis, Rodríguez et al. (2023) developed the **Chilean Dropout Early Warning System (EWS)**. The system processed 12 years of administrative data using the LightGBM gradient boosting algorithm to analyze individual student trajectories, identifying changes and accumulation of events that lead to school abandonment. This study firmly supports the proposed system's database architecture. By systematically routing current ENROLLMENT RECORDS into the HISTORICAL DATA repository at the end of each semester, the system ensures a continuous feedback loop that trains and recalibrates the predictive models for long-term institutional accuracy.

Local Studies

To address student attrition using local datasets, Deleña et al. (2025) developed a **Student Retention Predictive Model** that evaluated ten machine learning algorithms. By analyzing sociodemographic and academic factors, the model successfully identified at-risk students and improved timely graduation rates. This directly supports the data extraction logic of the proposed system's Process 4.0 (Generate Predictive Analytics and Forecasting). By mirroring the extraction of academic attributes from HISTORICAL DATA, the proposed system will deploy supervised learning algorithms to provide the Registrar Head with empirical attrition analytics rather than relying on reactive interventions.

Focusing on admission forecasting, Esquivel and Esquivel (2021) developed a **DSS Predictive Application** using Logistic Regression to predict undergraduate freshmen enrollment. The system achieved an accuracy rate of 80.5% in classifying students' enrollment decisions and utilized an interactive dashboard for data exploration. This validates the deployment of the forecasting engine inside the proposed system. By ingesting incoming profiles collected by the Admission Staff, the proposed system will similarly utilize predictive modeling to estimate enrollment demands, allowing the Program Chair to dynamically prepare section capacities in Process 1.0.

To streamline operations in Local Universities and Colleges (LUCs), Diaz and Jain (2025) developed the **BCC Automated Enrollment System**. The system integrated predictive and prescriptive analytics to replace outdated manual processes, significantly reducing administrative processing time. This study provides the overall architectural justification for the proposed system. Just as the BCC system centralized student registration, the proposed system will automate the fragmented 6-to-11 step physical routing, utilizing digital validation workflows to securely route data into ENROLLMENT RECORDS and HISTORICAL DATA.

Emphasizing the importance of early detection, a local study engineered an **Academic Performance Predictive Model** specifically for diploma students. Utilizing a Decision Tree algorithm, the system processed historical variables such as General Weighted Average (GWA) and absences to accurately identify at-risk learners. This directly maps to the proposed system's predictive attrition capabilities. By utilizing a similar algorithmic approach on STUDENT RECORDS, the proposed system ensures that the attrition analytics presented to the Dean and Registrar Head are mathematically grounded on localized academic constraints.

To optimize the evaluation of student applications, Jacobe and Pascua (2025) designed the **Applicant Tracking and Management System with Decision Support** using the Agile Scrum methodology. The system featured a priority alerts dashboard and structured profiling, achieving "Very Great Extent" compliance across all ISO/IEC 25010 software quality characteristics. This study provides the exact technical blueprint for the proposed system's Process 2.1 (Authenticate and Read Profile). Furthermore, it justifies the adoption of Agile methodology to iteratively build the proposed system and secure it against unauthorized access through strict Role-Based Access Control (RBAC).

Targeting the specific workflow congestions of manual academic advising, Almonteros et al. (2022) developed the **Student-Subject Encoding System**. The web application successfully emulated the curriculum-based evaluation capabilities of a faculty member by utilizing a rule-based algorithm to generate conflict-free schedules for irregular students. This study perfectly defends the core business logic of the proposed system's Process 2.2 (Validate Capacities and Prerequisites). Instead of faculty manually cross-referencing physical grade slips, the proposed system will dynamically cross-reference STUDENT RECORDS against CURRICULUM and SCHEDULING to automatically generate subject recommendations.

Investigating severe inefficiencies in inter-office workflows, Tabudlong and Estember (2024) analyzed and improved an **HEI Enrollment Portal**. The study revealed that unstructured scheduling and a lack of available rooms caused major queuing delays; hence, a centralized portal with dedicated student and faculty dashboards was implemented to organize course management. This directly supports the user interface architecture of the proposed system's Process 2.0. By providing a digital curriculum sheet and real-time section monitoring, the system will prevent scheduling overlaps and eliminate bureaucratic delays between the Registrar and Program Chair.

To centralize institutional data management, Grepon et al. (2021) implemented the **NBCC-School Management System**, a hybrid client-server application. The system securely managed student admissions, subject control, and grade assessments while utilizing a robust relational database framework. This study firmly supports the database schema and Data Layer of the proposed system architecture. By adopting a similarly centralized approach, the proposed system will ensure high data integrity, securely routing all transactions into

STUDENT RECORDS and ENROLLMENT RECORDS without the risk of data redundancy or loss.

Evaluating user behavior and system acceptability, a Philippine study assessed an **Online Enrollment System** using the Technology Acceptance Model (TAM). The research confirmed that Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) significantly dictate students' satisfaction and their behavioral intention to utilize digital portals. This study defends the Non-Functional Requirements (NFR) and User Design phase of the proposed system. By prioritizing an intuitive, high-fidelity web interface in the Presentation Layer, the proposed system ensures high student engagement and frictionless digital self-evaluation.

Finally, validating the importance of standardized software quality assurance, Lagman et al. (2025) evaluated the **Digital Academic Information System (DAIS)**. Using the ISO/IEC 25010 framework, the system was thoroughly audited by IT experts, achieving an "Excellent" rating in functional suitability, usability, and security. This provides the exact evaluation protocol required for Chapter 4 of the proposed system project. By subjecting the proposed enrollment and predictive analytics engine to the same strict ISO/IEC 25010 parameters, the researchers guarantee that the system will be highly reliable, secure, and ready for institutional deployment.

2.4 Synthesis of the Review of Related Literature and Studies

Based on the reviewed literature and related studies, there is overwhelming evidence supporting the transition from manual, paper-based academic processing to centralized web applications. Previous local studies have proven that web-based systems significantly improve transaction speeds, secure student data, and reduce the physical administrative burden on campus staff by eliminating the fragmented physical routing workflow.

However, a critical research gap remains evident within the Philippine higher education context. While local institutions have made commendable strides in adopting online registration portals, the vast majority of these platforms function only as static data repositories. They lack the intelligent features required to proactively assist academic leaders. Specifically, existing local systems do not integrate predictive analytics to forecast section demands, nor do they possess the automated rule-based logic required to map complex curriculum transitions. This technological gap directly mirrors the localized operational problems of Global Reciprocal Colleges (GRC), where the administration relies on unverified manual guesswork for faculty loading, resulting in dissolved sections and undetected student attrition.

To resolve this local problem (the "What"), the proposed system adopts the mathematical approaches and predictive methodologies proven effective in foreign studies (the "How"). Foreign research explicitly demonstrates that deploying supervised machine learning models—specifically **Random Forest** and **Extreme Gradient Boosting (XGBoost)**—yields exceptional accuracy in automating semester-based course planning and calculating dropout probabilities based on historical data.

The proposed **Automated Enrollment System with Predictive Analytics** directly addresses the local institutional gap by localizing these foreign-proven algorithms. By adopting the Random Forest and XGBoost frameworks to mine GRC's HISTORICAL DATA and STUDENT RECORDS, the proposed system will mathematically forecast incoming section capacities and generate empirical stuck-student reports. By synthesizing the immediate transactional benefits of a secure, local web-based enrollment portal with the strategic, long-term foresight provided by foreign predictive engines, this study delivers a comprehensive tool that empowers the Registrar, Program Chair, and Dean to execute proactive, data-driven institutional management.

To summarize this chapter, the review of related literature and studies establishes the technical and theoretical foundation for the proposed system. The explored literature highlighted the necessity of integrating machine learning algorithms, specifically Random Forest and Extreme Gradient Boosting (XGBoost), alongside robust relational database frameworks and strict Role-Based Access Control (RBAC) to secure institutional data. These theories provided the mathematical and infrastructural basis required to process complex educational datasets and enforce data access governance.

Furthermore, the reviewed foreign and local studies provided concrete architectural blueprints, proving that the transition from manual, paper-based routing to a centralized web application drastically reduces operational delays. By synthesizing these proven concepts and existing IT artifacts, the proposed system is architecturally justified to not only automate digital advising and prerequisite validation but also to proactively forecast section demands and student attrition. Ultimately, the reviewed works ensure that the development of the system is anchored on evidence-based methodologies, directly resolving the institution's localized enrollment workflow congestions.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Research Method

This study adopts the Agile Methodology as the primary software development approach to engineer the Automated Enrollment System with Predictive Analytics. To ensure systematic execution during the upcoming development phase, the researchers will utilize a structured Agile framework centered on Product Backlogs, Sprints, and Retrospectives. Initially, all complex system requirements—such as the digital curriculum mapping, automated prerequisite validation logic, and predictive forecasting algorithms—will be compiled and rigorously prioritized within the Product Backlog. The overall development lifecycle will then be divided into manageable, time-boxed iterations known as Sprints. During each future sprint, specific system modules extracted from the backlog will be systematically designed, built, and tested. Finally, at the conclusion of every sprint cycle, a Sprint Retrospective will be conducted with key institutional stakeholders, including the Registrar Head and Program Chair. This continuous retrospective phase ensures that every developed system increment is rigorously evaluated, allowing the researchers to rapidly adapt to evolving administrative constraints and flawlessly refine the enrollment business logic prior to full institutional deployment.

This figure 1 illustrates the iterative lifecycle phases of the Agile methodology utilized to systematically engineer and validate the Vantagepoint system modules through continuous sprint cycles.

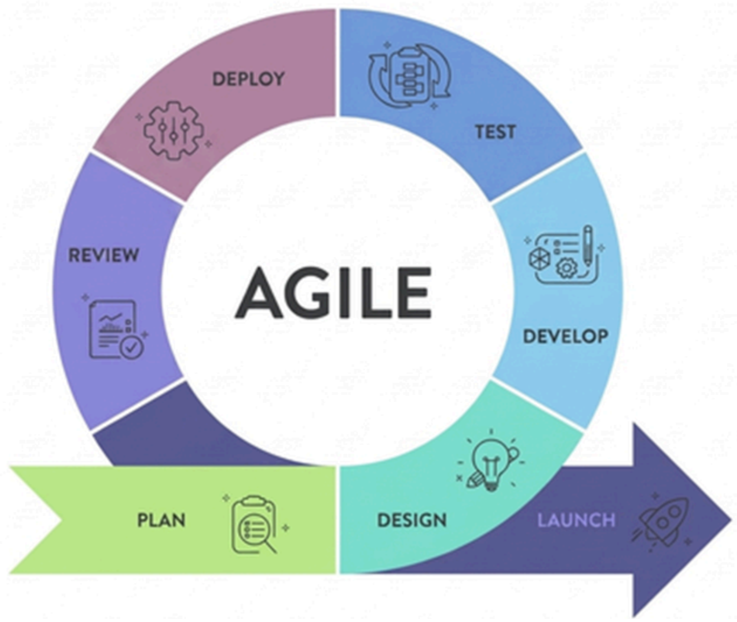

Figure 1. Agile Development Methodology

The development process follows iterative cycles where complex, interconnected modules—such as automated prerequisite validation and predictive forecasting—are prioritized in the Product Backlog. This framework is specifically selected for ***Global Reciprocal Colleges (GRC)*** *to address the high complexity of enrollment variables and administrative constraints identified during interviews with the* ***IT Department Program Chair****. By utilizing Agile, the research team can incrementally refine the software and adapt to newly discovered institutional rules during the actual software construction phase.*

Phase 1: Project Planning

During the Project Planning phase, the researchers identified the system goals, collected initial requirements, and prioritized features for the upcoming development sprints. This involved conducting interviews with key stakeholders at Global Reciprocal College, including the Program Chair of the IT Department, the Dean of the Academic Department, and the Registrar Head, to understand the existing manual enrollment process, pain points, and desired system features. The researchers also defined the technology stack, which consists of Next.js 16 with React for the frontend, Laravel as the RESTful API backend, MySQL as the database management system, and Laravel Sanctum for API token-based authentication. From the gathered requirements, the system boundary was established through the Context Diagram, which depicts the interaction between external entities and the proposed system.

The Context Diagram establishes the absolute system boundaries and global data interactions between the centralized platform and exactly nine (9) distinct external entities.

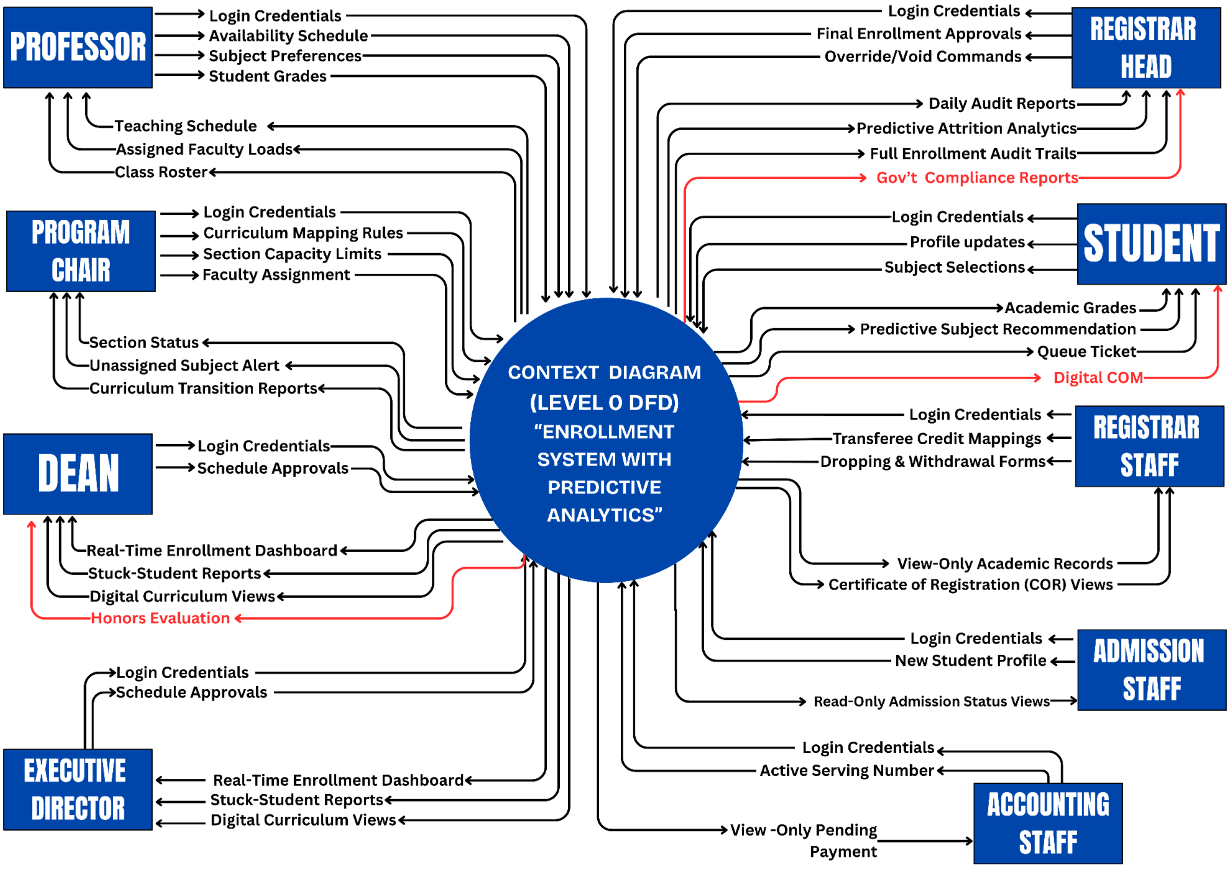

Figure 2 maps the global exchange of noun-based data packets, ensuring that all complex academic logic is systematically bounded by predefined institutional data flows. To address the panel's requirements for reporting and student portal expansion, the interactions are updated as follows:

- **Student:** Injects Login Credentials, **User Account Updates**, and Subject Selections; in return, receives Academic Grades, Predictive Subject Recommendations, the digital Queue Ticket, and the **Digital Certificate of Matriculation (COM)**.
- **Admission Staff:** Injects Login Credentials and New Student Profiles to initiate user accounts, receiving Read-Only Admission Status Views.
- **Professor:** Submits Availability Schedules, Subject Preferences, and Encoded Final

Grades, while receiving Teaching Schedules, Assigned Faculty Loads, and Class Rosters.

- **Program Chair:** Injects Curriculum Mapping Rules, Section Capacity Limits, and Faculty Assignments; receives Section Status updates, Unassigned Subject Alerts, and **[NEW] Curriculum Transition Reports** to optimize sectioning logic.
- **Dean:** Submits preliminary Schedule Approvals and receives the Real-Time Enrollment Dashboard, Stuck-Student Reports, Digital Curriculum Views, and **Honors Evaluation / Dean's List Reports**.
- **Executive Director:** Injects final Schedule Approvals to authorize system parameters, receiving institutional analytics through the Real-Time Enrollment Dashboard and Digital Curriculum Views.
- **Registrar Head:** Injects Final Enrollment Approvals and Override Commands; receives Daily Audit Reports, Predictive Attrition Analytics (covering Drops, Withdrawals, and Failures), and **Gov't Compliance Reports (e.g., CHED Enrollment List)**.
- **Registrar Staff:** Inputs Transferee Credit Mappings and Dropping/Withdrawal Forms, receiving View-Only Academic Records and Certificate of Registration (COR) Views.
- **Accounting Staff:** Inputs the Active Serving Number and receives View-Only Pending Payment status to validate the student's financial clearance sequence.

Figure 3 illustrates the structural decomposition of the core system into four (4) major action-oriented subsystems, now featuring an integrated **Closed-Loop Forecasting Feedback and [NEW] Administrative Compliance reporting module**

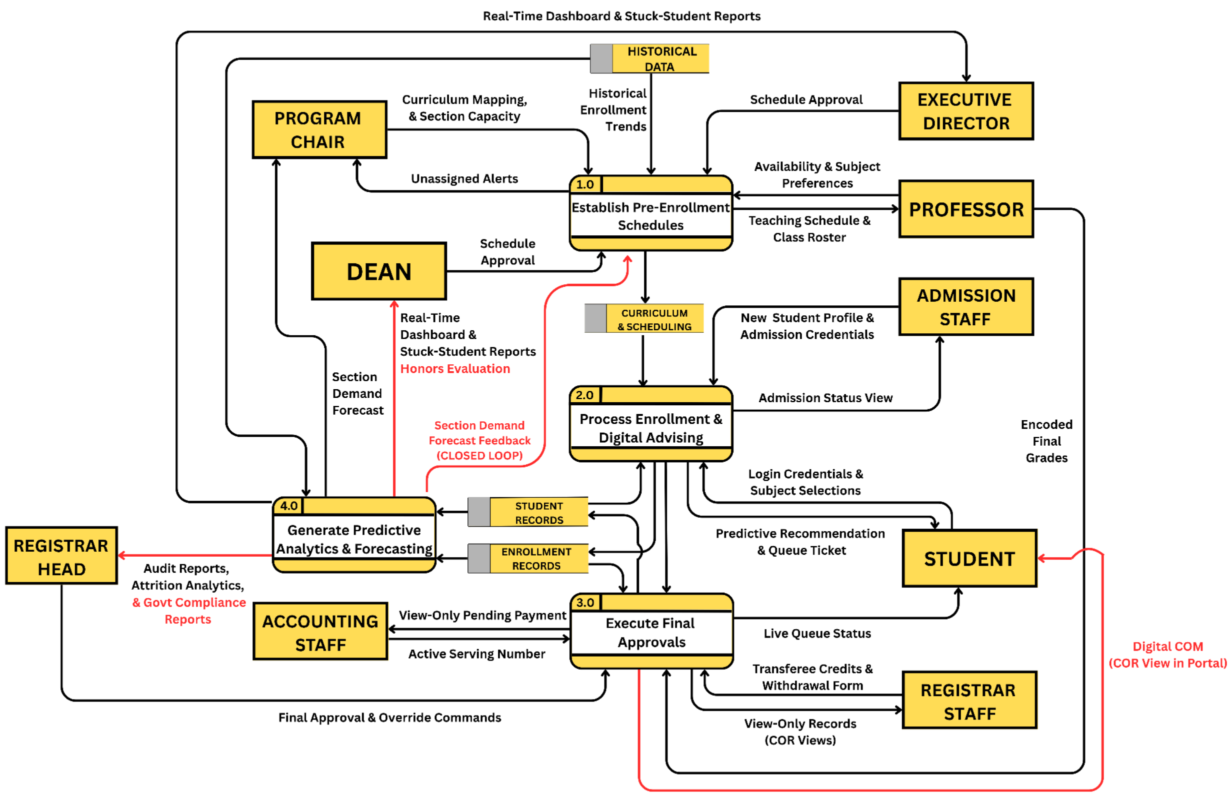

Figure 3. DFD Level 1 of Proposed System

Figure 3 details the data transitions between four primary processes, ensuring absolute data traceability across the institution's localized data stores. To address the panel's requirements for end-to-end reporting and student portal transparency, the processes are refined as follows:

- **Process 1.0: Establish Pre-Enrollment Schedules:** This process handles the academic logic required before the enrollment period begins. It receives Curriculum Mapping and Section Capacity from the Program Chair, and Availability and Subject Preferences from the Professor. It now ingests the **Section Demand Forecast Feedback** directly from Process 4.0, creating a **Closed-Loop Mechanism** that allows the Program Chair to adjust capacities based on empirical predictions. Once locked via the Dean and Executive Director's approval, finalized schedules are stored in CURRICULUM and SCHEDULING.

- **Process 2.0: Process Enrollment and Digital Advising:** This module automates the core registration and rule-based advising workflow. It captures New Student Profile and Admission Credentials from the Admission Staff to **initialize User Accounts**. For existing scholars, it cross-references academic history in STUDENT RECORDS against curriculum rules to output Predictive Recommendations and the digital Queue Ticket.
- **Process 3.0: Execute Final Approvals:** This process governs record modifications and financial hand-offs. It ingests Transferee Credits and Withdrawal Forms from the Registrar Staff and executes final administrative clearance via the Registrar Head. Upon successful financial validation from the Accounting Staff, the system now automatically generates and routes the **Digital Certificate of Matriculation (COM)** directly to the Student's portal view.
- **Process 4.0: Generate Predictive Analytics and Forecasting:** Functioning as the system's intelligent core, this process extracts current registration volumes and academic statistics. It now executes three critical reporting sub-routines: (1) **Attrition Analytics** (calculating probabilities for Drops, Withdrawals, and Failures for the Registrar Head), (2) **Honors Evaluation** (identifying Dean's List candidates for the Dean), and (3) **Gov't Compliance Reports** (generating mandatory enrollment documentation for the Registrar Head). Processed data is routed back into HISTORICAL DATA and the **Section Demand Forecast Feedback loop** to recalibrate future models.

This diagram details the specific computational sub-routines and multi-level approval workflows required to define curriculum rules and faculty assignments, now enhanced with a **Closed-Loop Predictive Feedback mechanism** to optimize institutional sectioning.

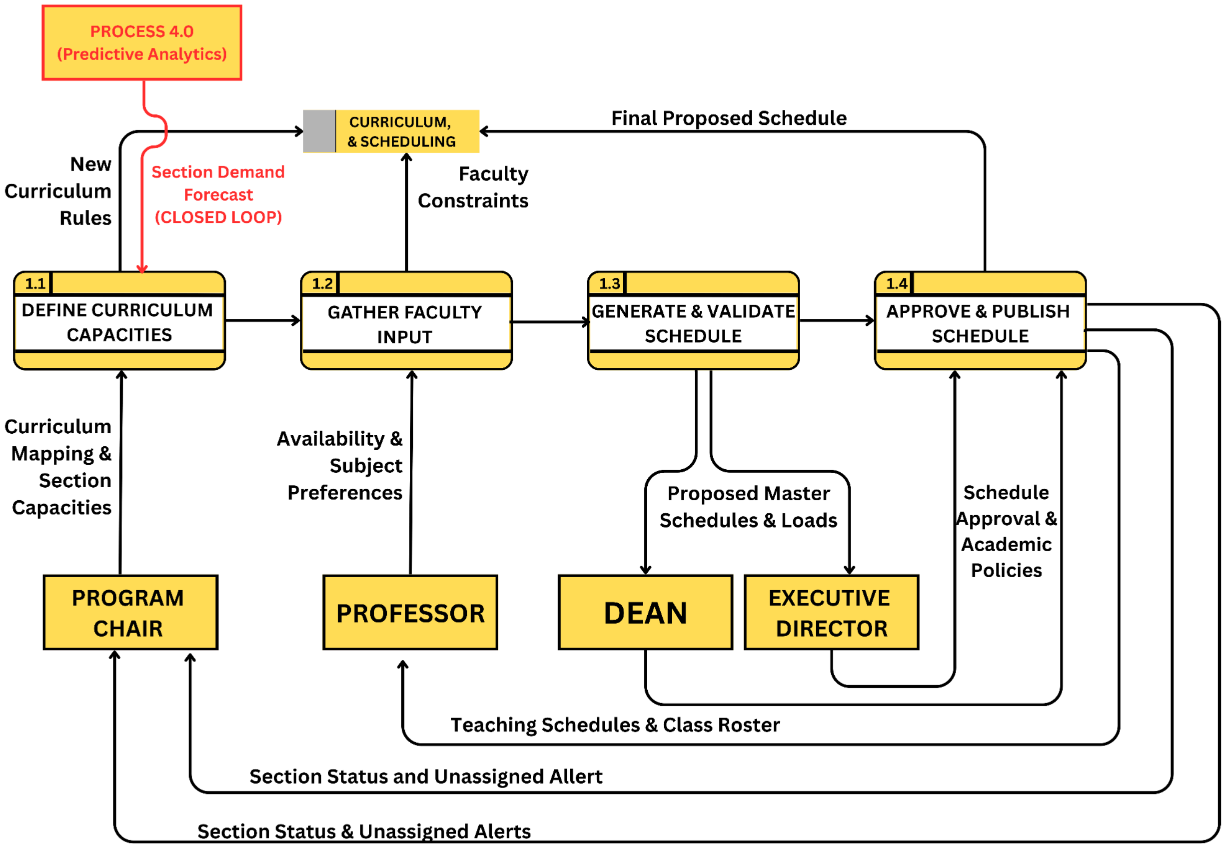

Figure 4. DFD Level 2 of Establish Pre-Enrollment Schedules

Process 1.0 focuses on setting the system's mathematical baseline through four sequential steps. **[UPDATED]** The workflow now integrates empirical forecasting data to ensure that class section offerings are mathematically aligned with student demand.

- **Process 1.1: Define Curriculum Capacities:** Captures the Curriculum Mapping and Section Capacities injected by the Program Chair. This process now ingests the **Section Demand Forecast (Closed Loop)** from Process 4.0, allowing the system to algorithmically suggest optimized section counts based on historical trends before logging the New Curriculum Rules into the data store.

- **Process 1.2: Gather Faculty Input:** Captures the Availability and Subject Preferences injected by the Professor. The system processes these into validated Faculty Constraints and routes them to the data store.
- **Process 1.3: Generate and Validate Schedule:** Extracts compiled curriculum rules and faculty constraints to map out conflict-free class allocations. It outputs Proposed Master Schedules and Loads directly to the Dean.
- **Process 1.4: Approve and Publish Schedule:** Ingests the Schedule Approval and Academic Policies from the Dean, followed by authorization from the Executive Director. The system routes the Final Proposed Schedule into the data store, triggering the distribution of Teaching Schedules and Class Rosters to the Professors, and Section Status and Unassigned Alerts to the Program Chair.

This figure details the automated sub-routines responsible for digital student evaluation, rule-based prerequisite validation, and the generation of the digital queue ticket.

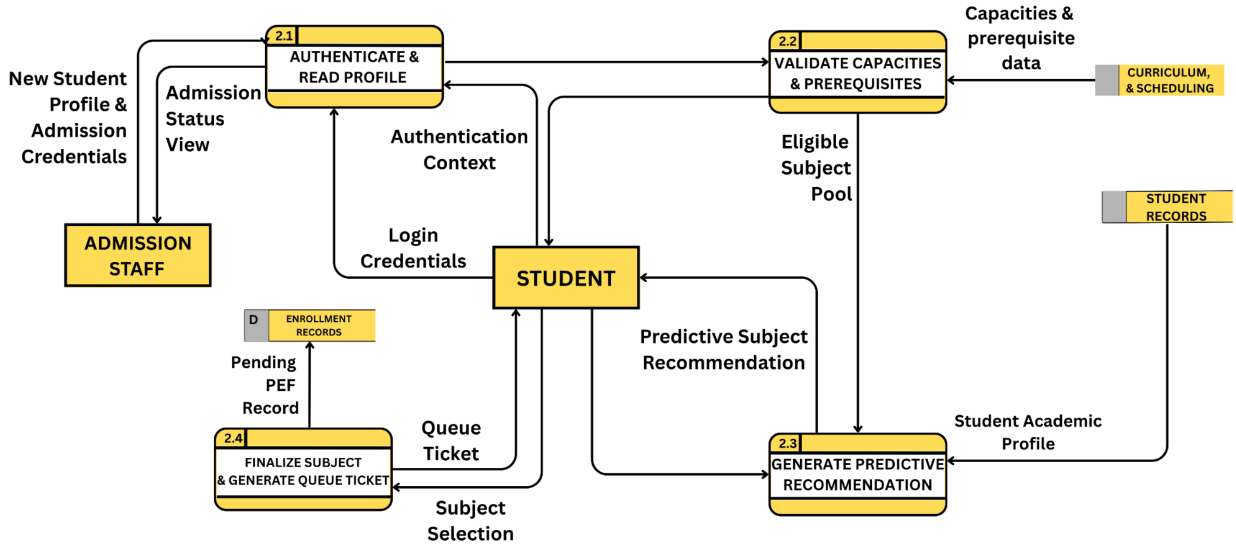

Figure 5. DFD Level 2 of Enrollment and Digital Advising

(Description Below): Process 2.0 automates the registration cycle by executing **Authenticate and Read Profile (2.1)** followed by **Validate Capacities and Prerequisites (2.2)**, where student records are cross-referenced against curriculum constraints. The system outputs a **Predictive Subject Recommendation (2.3)** to guide subject selection and concludes with **Finalize Subject and Generate Queue Ticket (2.4)**, which logs the pending transaction into the enrollment records data store.

- Process 2.1: Authenticate and Read Profile: Captures the New Student Profile and Admission Credentials from the Admission Staff to output an Admission Status View. It processes Login Credentials injected by the Student, outputting the validated Authentication Context.
- Process 2.2: Validate Capacities and Prerequisites: Receives the Authentication Context to trigger data retrieval. It extracts Fetch Student Data from STUDENT RECORDS and

cross-references it against Capacities and Prerequisite Data from CURRICULUM and SCHEDULING, outputting an Eligible Subject Pool.

- Process 2.3: Generate Predictive Recommendation: Ingests the Eligible Subject Pool alongside the Student Academic Profile. It formats a rule-compliant academic load, outputting the Predictive Subject Recommendation directly to the Student.
- Process 2.4: Finalize Subject and Generate Queue Ticket: Captures the final Subject Selection verified by the Student. The system routes the Pending PEF Record into ENROLLMENT RECORDS and outputs the Queue Ticket to the Student.

This diagram illustrates the processing logic for Process 3 (Execute Final Approvals), now enhanced with **Digital Certificate of Matriculation (COM) generation** and **[NEW] Attrition Data routing** to ensure a complete enrollment-to-certification cycle.

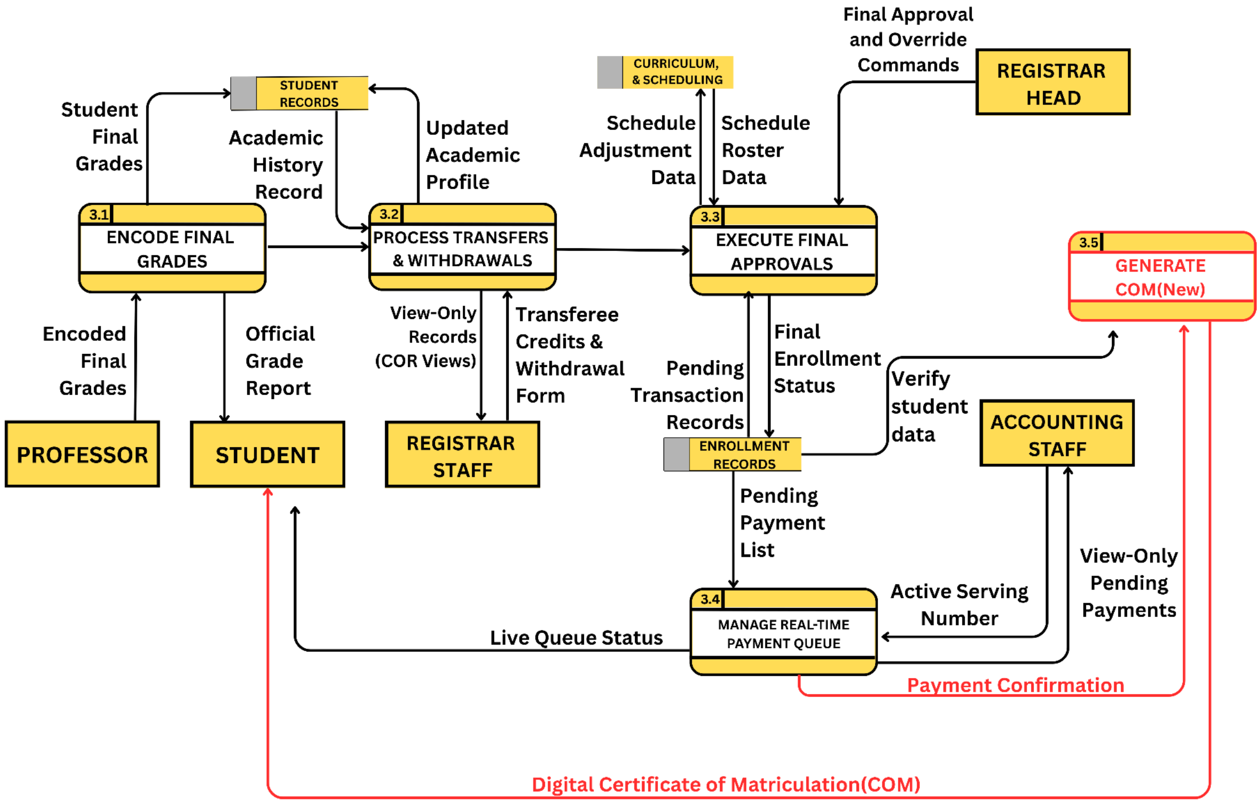

Figure 6. DFD Level 2: Decomposition of Process 3 (Execute Final Approvals)

Figure 6 structurally decomposes the final validation phase into five key sub-processes: Encode Final Grades (3.1), Process Transfers and Withdrawals (3.2), Execute Final Approvals (3.3), Manage Real-Time Payment Queue (3.4), and **Generate COM (3.5)**. This module bridges the academic and financial departments of GRC, ensuring student records are securely locked and certified only after successful financial clearance.

- **Process 3.1: Encode Final Grades:** Securely captures the Encoded Final Grades injected by the Professor. The system directly routes these grades into STUDENT RECORDS, providing the necessary data for **Attrition Metrics (Failures)** and **Honors Evaluation** in Process 4.0.

- **Process 3.2: Process Transfers and Withdrawals:** Captures Transferee Credits and Withdrawal Forms from the Registrar Staff. This process routes **Withdrawal and Drop data** to the analytics module to support the panel's requirement for detailed **Attrition Reporting**.
- **Process 3.3: Execute Final Approvals:** Ingests the Updated Academic Profile and Schedule Roster Data. The Registrar Head injects Final Approval and Override Commands to compute assessments and push the Final Enrollment Status into ENROLLMENT RECORDS.
- **Process 3.4: Manage Real-Time Payment Queue:** Extracts the Pending Payment List and routes the Active Serving Number to the Accounting Staff. Upon receiving **Payment Confirmation**, the process now triggers the **Generate COM (3.5) sub-routine** to finalize the student's portal documentation.
- **Process 3.5: Generate COM:** This newly integrated process ingests the Payment Confirmation and validated enrollment data to produce the **Digital Certificate of Matriculation (COM)**. The artifact is directly routed to the **Student Portal**, providing immediate official proof of enrollment as requested by the panel.

This figure represents the system's analytical core, now expanded into six (6) distinct sub-processes to provide high-fidelity attrition metrics, automated honors evaluation, and mandatory institutional compliance reporting.

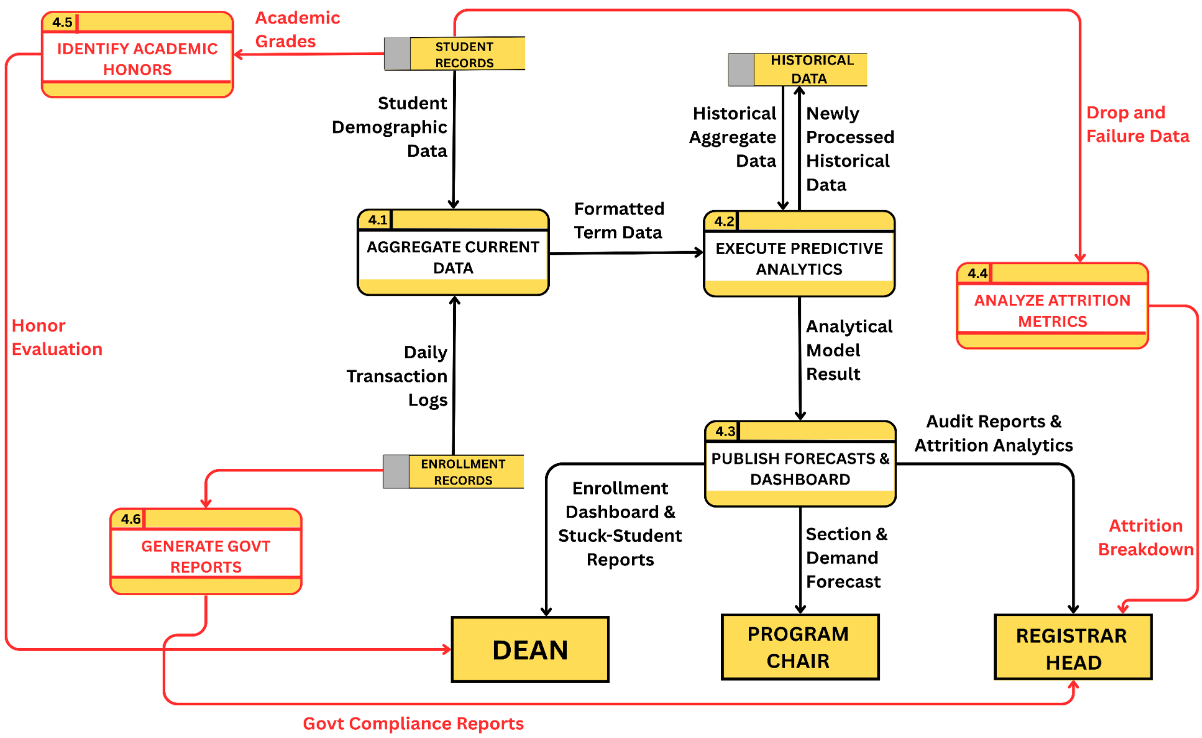

Figure 7. DFD Level 2: Decomposition of Process 4

(Generate Predictive Analytics and Forecasting)

Process 4.0 functions as the system's intelligent layer by executing Aggregate Current Data (4.1) and Execute Predictive Analytics (4.2) on information pulled from STUDENT RECORDS and HISTORICAL DATA. **To address the panel's requirements for granular oversight, the module now includes dedicated routines for Attrition Analysis (4.4), Honors Identification (4.5), and Government Reporting (4.6)**. The results are finalized in Publish Forecasts and

Dashboard (4.3), while raw transaction volumes are routed back into the feedback loop to recalibrate future models.

- **Process 4.1: Aggregate Current Data:** Systematically extracts Student Demographic Data from STUDENT RECORDS and Daily Transaction Logs from ENROLLMENT RECORDS to output highly structured Formatted Term Data.
- **Process 4.2: Execute Predictive Analytics:** Ingests the Formatted Term Data and strictly cross-references it with Historical Aggregate Data. It routes newly processed term statistics back into HISTORICAL DATA to train future models, outputting the Analytical Model Result.
- **Process 4.3: Publish Forecasts and Dashboard:** Ingests the Analytical Model Result and translates it into institutional reports. It routes the **Section and Demand Forecast** to the Program Chair to drive the closed-loop sectioning logic in Process 1.1, while providing the Enrollment Dashboard and Stuck-Student Reports to the Dean and Executive Director.
- **Process 4.4: Analyze Attrition Metrics:** This process extracts Drop and Failure Data from the student data repository. It generates a detailed **Attrition Breakdown** (categorizing students based on drop, withdrawal, and failure probabilities) for the Registrar Head, directly addressing the panel's requirement for comprehensive attrition tracking.
- **Process 4.5: Identify Academic Honors:** Pulls Academic Grades from STUDENT RECORDS to mathematically evaluate performance against institutional criteria. It outputs the **Honor Evaluation** report for the Dean to facilitate the automated identification of Dean’s Listers and honor candidates.
- **Process 4.6: Generate Govt Reports:** Ingests Daily Transaction Logs to format mandatory documentation for external regulatory bodies. It outputs the **Govt Compliance**

Reports (e.g., CHED enrollment lists) to the Registrar Head, ensuring institutional compliance without manual data compilation.

Table 1. Planned Sprint Breakdown for System Architecture and Design

Sprint	**Target DFD Subsystem**	**Key Design and Architectural Deliverables**
Sprint 1	Entity and Access Control Mapping	Wireframe the secure authentication interfaces for all nine (9) external entities. Design the Role-Based Access Control (RBAC) middleware logic to enforce strict data boundaries among the Student, Program Chair, Dean, Executive Director, Professor, Admission Staff, Registrar Head, Registrar Staff, at Accounting Staff.
Sprint 2	Process 1: Establish Pre-Enrollment Schedules	Map the curriculum logic and section capacity rules. Design the UI for the Professor to submit availability and subject preferences. Wireframe the UI for the Program Chair to draft faculty loads. Architect the multi-level schedule approval workflow routing from the Dean to the Executive Director for final locking before storing the exact data packet to CURRICULUM and SCHEDULING.

Sprint 3	Process 2: Process Enrollment and Digital Advising	Design the automated prerequisite validation algorithms. Create the logical flow for fetching student histories from STUDENT RECORDS, cross-referencing them against CURRICULUM and SCHEDULING to output the Predictive Subject Recommendation to the Student.
Sprint 4	Process 2.4: Finalize Subject and Generate Queue Ticket	Wireframe the digital Pre-Enrollment Form (PEF). Design the hand-off logic where the system outputs the Digital Queue Ticket to the student and routes the exact Active Serving Number to the Accounting Staff's interface.
Sprint 5	Process 3: Manage Academic Records and Approval	Wireframe the Faculty Grading interface for professors to encode final grades. Design the Registrar Head's overriding dashboard. Wireframe the interface for the Registrar Staff to input Transferee Credits and Withdrawal Forms. Design the Accounting Staff's view-only interface for verifying the Queue Ticket.
Sprint 6	Process 4: Generate	Draft the statistical algorithms (e.g., linear regression) for demand forecasting based on historical trends. Design the

	Predictive Analytics and Forecasting	critical Feedback Loop schema, ensuring data from D: ENROLLMENT RECORDS routes back into D: HISTORICAL DATA for the Registrar Head's Predictive Attrition Analytics. Draft the statistical algorithms (e.g., linear regression) for demand forecasting based on historical trends. Design the critical Feedback Loop schema, ensuring data from ENROLLMENT RECORDS routes back into HISTORICAL DATA for the Registrar Head's Predictive Attrition Analytics.
Sprint 7	System Prototyping and Balancing Verification	Consolidate all module designs into a high-fidelity prototype. Execute a strict system balancing audit to verify that all interface inputs and outputs perfectly match the arrows in the Level 0 Context Diagram, Level 1 DFD, and Level 2 DFD.

Phase 3: Develop

During the Develop phase, the researchers will initiate the actual software construction of the Automated Enrollment System with Predictive Analytics. Following the blueprints established in the Data Flow Diagrams (DFDs) and Entity-Relationship Diagram (ERD), the frontend interface will be programmed using Next.js 16 and React to ensure a highly responsive user experience. Simultaneously, the server-side business logic and RESTful APIs will be built using the Laravel framework, secured by Laravel Sanctum for token-based authentication. Crucially, during this phase, the predictive analytics engine will be coded and integrated into the backend. The researchers will script the machine learning algorithms—specifically Random Forest and XGBoost—to accurately fetch and process data from the centralized MySQL database, specifically targeting the mathematical extraction of parameters from HISTORICAL DATA and STUDENT RECORDS.

Phase 4: Test

In the Test phase, the developed system increments will undergo rigorous quality assurance and debugging before being presented to the end-users. The researchers will conduct Unit Testing to verify individual module functions, such as the automated prerequisite validation, and Integration Testing to ensure that the data flawlessly routes between the Registrar, Accounting, and Academic department interfaces. Furthermore, the predictive analytics engine will be subjected to statistical validation using standard error metrics to guarantee the precision of its section demand forecasting. Finally, a User Acceptance Test (UAT) will be conducted with the key stakeholders of Global Reciprocal Colleges (GRC), evaluating the system strictly against the ISO/IEC 25010 software quality standards to measure its functional suitability, performance efficiency, usability, and security.

Phase 5: Deploy

Once the system passes all testing parameters and secures stakeholder approval, the Deploy phase will commence. The web-based system will be hosted on a secure local or cloud-based server environment to make it fully accessible to the nine (9) authorized external entities. During this implementation stage, the centralized database will be populated with initial institutional data, including the digitized curriculum mapping and active faculty profiles. The Role-Based Access Control (RBAC) will be strictly enforced, ensuring that students, professors, and administrators can securely log in and access their respective bounded dashboards without encountering data overlapping or unauthorized exposure.

Phase 6: Review

In the final Agile phase, the researchers will conduct a comprehensive Sprint Retrospective to assess the overall performance of the deployed system. Feedback will be continuously gathered from the Registrar Head, Program Chair, and Dean regarding the system's efficiency in eliminating the physical enrollment routing and its accuracy in predicting faculty loading demands. Any identified minor bugs or logical constraints will be documented in the product backlog for immediate maintenance updates. Moreover, this phase will initialize the continuous feedback loop of the AI engine, ensuring that newly validated transaction records from ENROLLMENT RECORDS are successfully routed back into HISTORICAL DATA to recalibrate and train future predictive models.

3.2 Conceptual Framework

The Input-Process-Output (IPO) model provides a high-level logical summary of the system's overall architecture, serving as a conceptual translation of the Context Diagram and DFD Level 1

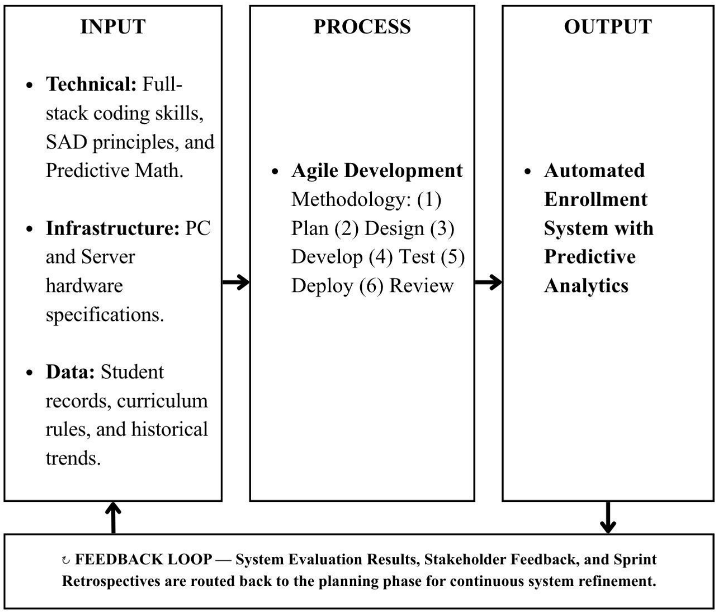

Figure 8. Input Process Output Model of the Proposed System

Figure 8 Framework of the "Automated Enrollment System with Predictive Analytics" utilizes the Input-Process-Output (IPO) model to provide a high-level logical summary of the project's software development lifecycle. Instead of merely mapping system data, this model serves as the conceptual roadmap for engineering the software, ensuring strict traceability from the initial requirements gathering to the final system deployment.

- **Input:** The foundational requirements and resources necessary for the systematic development of the software. This strictly includes the technical proficiency of the researchers (Full-stack coding, Systems Analysis and Design principles, and predictive mathematics), the necessary hardware and server infrastructure, and the raw institutional data (student records, curriculum rules, and historical trends) required to train the predictive machine learning models.

- **Process:** The transformation of the gathered inputs into the final system architecture will strictly utilize the Agile Development Methodology. This methodological process dictates the systematic engineering of the software through iterative sprint cycles divided into six core structural phases: (1) Project Planning and requirements gathering, (2) System Design through architectural modeling, (3) Development of the web application and predictive algorithms, (4) Testing and software quality assurance, (5) Deployment, and (6) Stakeholder Review and retrospectives.
- **Output:** The definitive and tangible output of this methodological development process is the fully engineered **"Automated Enrollment System with Predictive Analytics"**.
- **Feedback Loop:** The continuous cycle of evaluation and refinement intrinsic to the Agile framework. System testing results, stakeholder feedback from the Dean and Registrar, and Sprint Retrospective findings are systematically routed back to the project planning phase. This ensures that any architectural bugs or changing institutional requirements are continuously addressed in subsequent development iterations.

3.3 Data Gathering Tools and Procedures

To plan the project and list the tasks for the Agile Sprint Planning, the researchers used different data gathering methods. This helped the team clearly understand the current manual steps, processing delays, and school data rules at Global Reciprocal Colleges (GRC).

Interviews

Video-call interviews were done with key school staff. Their answers were used as the main guide to define user roles and list down the system features needed for the sprint backlogs:

- **Program Chair of the IT Department:** Explained how schedules and faculty loads are made, and the challenges of handling irregular students and changes in the subjects.

- **Dean of the Academic Department:** Highlighted the need for a central digital copy of the curriculum, automated checking of prerequisites, and separating clearances from the actual enrollment.
- **Registrar Head:** Pointed out slow processes like the manual checking of subjects, repetitive signing of clearances, and the slow searching of physical records. This delay is particularly severe for returning and transferee students, whose legacy records must be manually verified at the Admission Office and cross-referenced by the Academic Advisers before they are allowed to proceed to the core enrollment routing.

Observation

To empirically quantify the physical routing delays, the researchers executed a formal Time-and-Motion Study during peak enrollment days. Rather than passive observation, the development team systematically tracked the exhaustive physical movements of the students across different administrative nodes. This method mathematically measured the waiting times and transit durations as students navigated between the 3rd-floor Academic Advising rooms down to the 2nd-floor Computer Laboratory, and across the Clinic, Library, Registrar, Cashier, and Admission Office. The empirical data gathered from this study provided the exact operational baseline required to justify the severe processing delays caused by manual document hand-offs.

Document Analysis

To properly design the system's database, the researchers reviewed available blank school templates and standard curriculum sheets. Due to strict institutional data privacy rules enforced by the Registrar's Office, actual student records and filled-out Pre-Enrollment Forms (PEF) were not collected. Instead, the team verified the required data fields during the interviews to ensure the new database matches the school's official format without violating data security.

Procedures

Based on the synthesized data from the interviews and the Time-and-Motion Study, the researchers employed Business Process Analysis through formal Workflow Mapping to architecturally document the existing manual enrollment setup. This systematic mapping mathematically verified the exact number of transactional steps executed by the students. For old and regular students, the workflow successfully mapped a strict six-step routing. However, for new and transferee students, the mapping explicitly revealed an exhaustive eleven-step (11-step) architecture due to the fragmented admission and medical prerequisites required before entering the core enrollment cycle.

By translating these findings into a Master Activity Flowchart (Figure 9), the researchers established the exact technical justification for the operational congestions declared in this study. The mapped workflow strictly highlights the manual verification loop of Pre-Enrollment Forms (PEF), proving that human intervention and inter-floor physical document routing cause the most significant institutional administrative chokepoints.

1. **Secure official permission** from the GRC administration to conduct a Time-and-Motion Study on their current system.
2. **Interview** the Program Chair, Dean, and Registrar Head to understand the institutional academic rules and constraints.

3. **Execute Workflow Mapping** by tracing the exact physical steps of old, new, and transferee students during peak enrollment.
4. **Review blank institutional templates** to identify the exact data fields required for the centralized database schema.
5. **Synthesize all gathered data** into actionable Agile sprint backlogs and architect the system boundaries using Context Diagrams and Data Flow Diagrams.

This flowchart maps the institutional manual enrollment architecture of **Global Reciprocal Colleges** to highlight physical routing bottlenecks and manual verification delays.

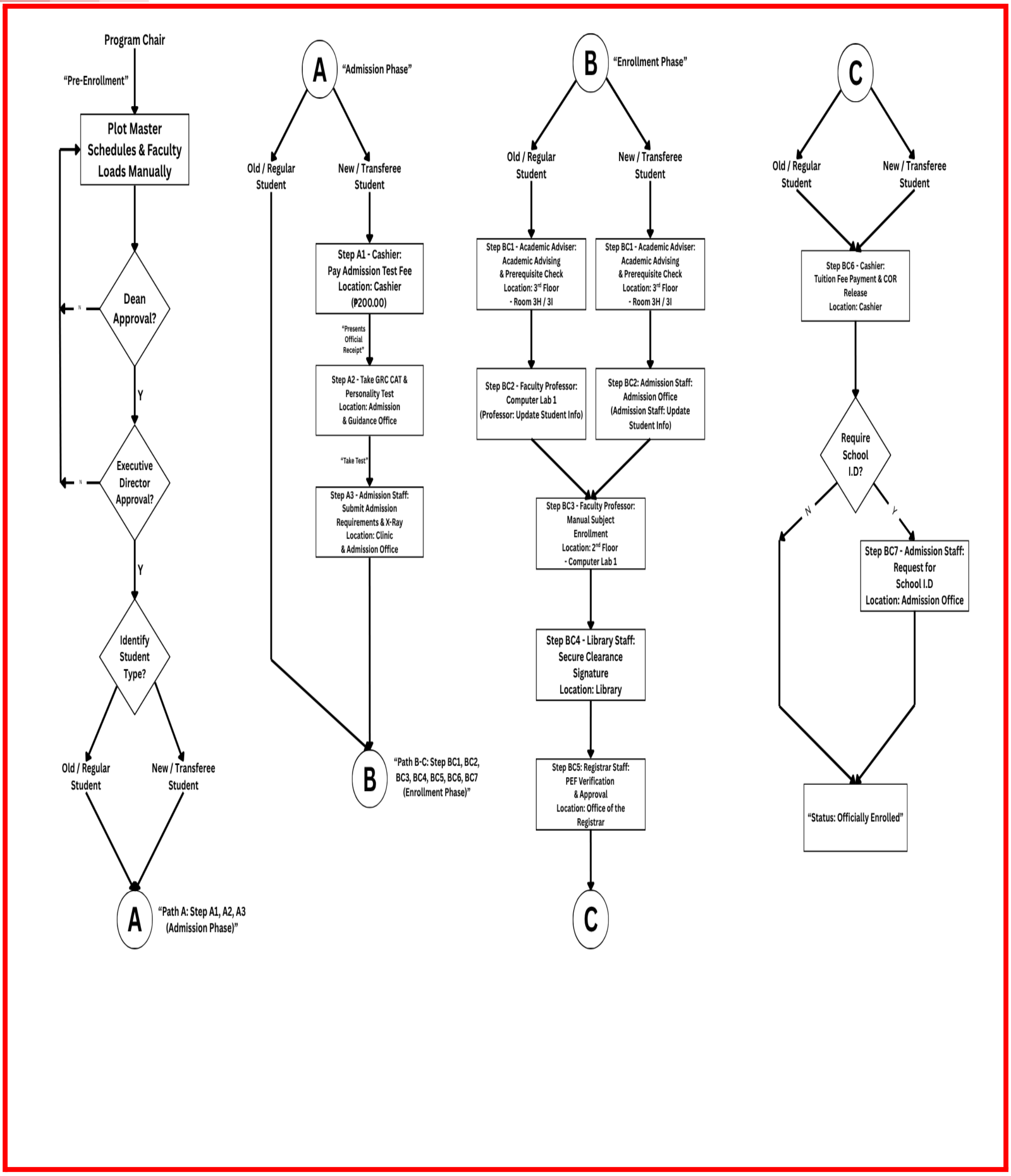

Figure 9. Process Flowchart of the Existing Enrollment System

Figure 9 illustrates the streamlined, end-to-end digital architecture of the proposed system, structurally mapping the complete student lifecycle from initial registration up to institutional compliance. Unlike the fragmented manual setup, this automated workflow systematically executes the following sequence:

1. **Digital Advising and Prerequisite Validation:** The system cross-references the student's academic history against institutional rules.
2. **AI-Driven Sectioning and Enlistment:** The predictive model evaluates historical data to allocate students into optimal class schedules, strictly opening sections only when forecasted demand meets the institutional threshold.
3. **Assessment and Payment Bridging:** The system generates a digital queue ticket and synchronizes the real-time financial assessment with the Accounting dashboard.
4. **Official Enrollment Confirmation:** Upon payment validation, the system instantly provisions the Digital Certificate of Matriculation (COM) directly to the student portal.
5. **Automated Government Report Generation:** Concluding the workflow, the system's analytical core (Process 4.0) aggregates all locked enrollment transactions to automatically formulate and output mandatory regulatory documents (e.g., CHED Enrollment Lists) for the Registrar Head.

3.4 Project Design

This section details the structural design of the "Automated Enrollment System with Predictive Analytics," encompassing the system's physical architecture, behavioral user interactions, absolute data boundaries, and logical database schemas

To ensure absolute logical consistency across all system models, the researchers implemented strict Data Traceability and Architectural Balancing. Every noun-based data packet identified in the

Level 0 Context Diagram—such as the *New Student Profile*, *Subject Selections*, and *Encoded Final Grades*—is systematically mapped and cascaded through the Level 1 and Level 2 Data Flow Diagrams without any arbitrary renaming or data loss. Furthermore, these exact data flows act as the structural bridge to the system's Data Layer, serving as the direct inputs that populate the Third Normal Form (3NF) tables in the Entity-Relationship Diagram (ERD). For instance, the *New Student Profile* data flow from the Admission Staff is strictly bound to the Student_Profile entity, while the *Login Credentials* logically map to the centralized System_Users table. This rigid traceability protocol, governed by the system's Data Dictionary, guarantees that there are no orphaned data flows, missing database entities, or logical disconnects between the behavioral boundaries and the relational database schema of the proposed system.

3.4.1 System Architecture

This figure illustrates the three-tier Client-Server architecture of the Vantagepoint system, comprising the Presentation, Application, and Data layers for secure and scalable processing.

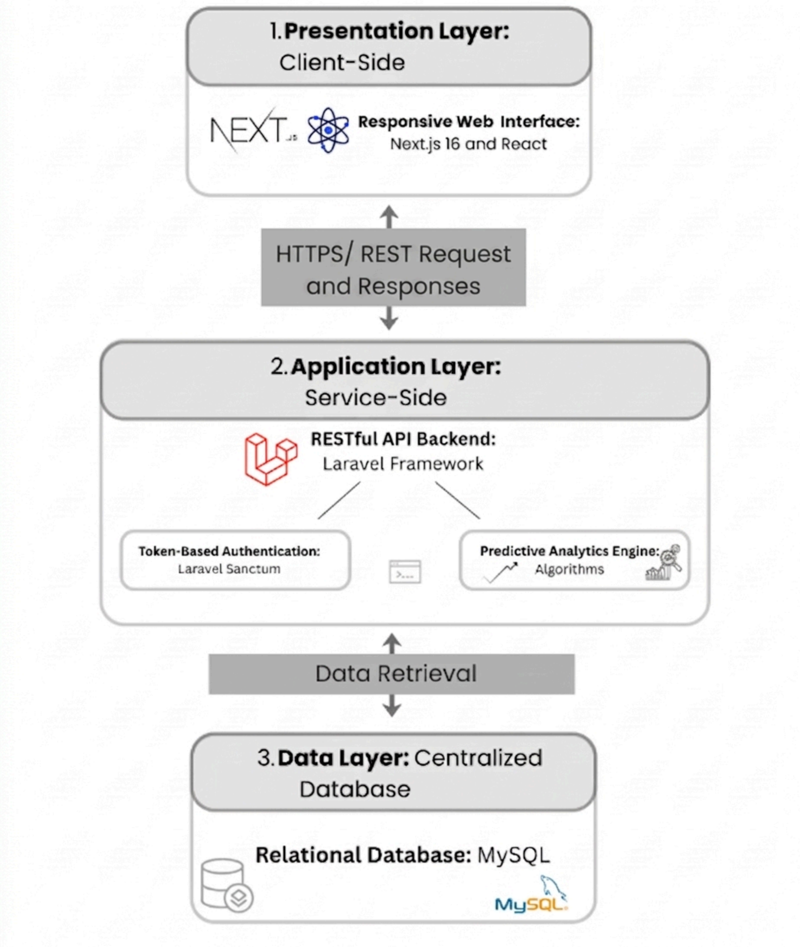

Figure 10. System Architecture of the Proposed System

Figure 10 shows the proposed system employs a modern Client-Server architecture designed for high scalability and secure data processing. As illustrated in the System Architecture Diagram, the application is divided into three interconnected layers: the Presentation Layer (Next.js/React frontend), the Application Layer (Laravel RESTful API), and the Data Layer (MySQL database) handling the predictive models.

The Use Case Diagram maps the behavioral interactions and Role-Based Access Control (RBAC) architecture between the system and its nine (9) authorized external actors.

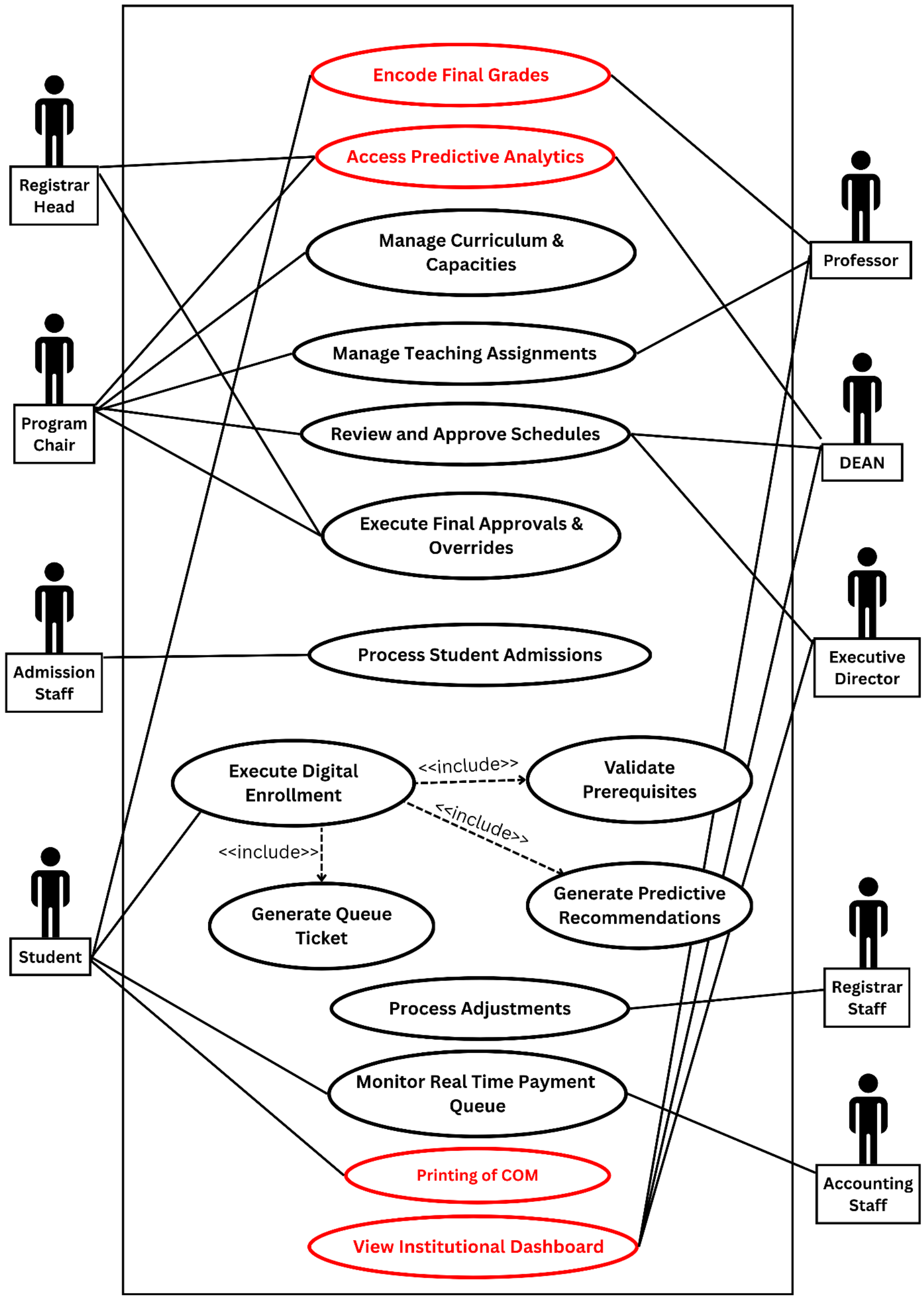

Figure 11. Use Case Diagram of the Proposed System

Figure 11 Use Case Diagram maps the behavioral interactions and Role-Based Access Control (RBAC) architecture between the Vantagepoint system and its nine authorized external actors, **integrating new student certification and administrative compliance modules as requested by the panel.** To ensure seamless inter-user visibility, the operational workflow is defined as follows:

- The workflow begins with the Admission Staff, who verify credentials and initiate new profiles, enabling the Student to remotely **manage their personal account**, select subjects, view academic grades, receive predictive recommendations, generate a queue ticket, and **execute the Printing of COM (Certificate of Matriculation)** upon successful financial clearance.
- On the academic side, the Program Chair controls curriculum mapping rules, sets section capacities, and drafts faculty assignments, while the Professor submits schedule availability and **securely encodes final grades**—an output that directly populates the academic progress view of other users.
- Administrative oversight is enforced by the Dean and Executive Director, who review schedules and monitor the **View Institutional Dashboard**, which now includes **Honors Evaluation (Dean's List)** and real-time enrollment analytics for strategic decision-making.
- For institutional records management, the Registrar Head utilizes the **Access Predictive Analytics** module to monitor attrition (including drops, withdrawals, and failures) and **generate mandatory Gov't Compliance Reports**. The Registrar Staff supports this by processing transferee credits and withdrawal forms.
- The registration cycle officially concludes with the Accounting Staff, who manage the real-time payment queue to finalize the student's financial clearance, effectively triggering the availability of the **Digital COM** for the student.

3.4.3 Entity Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) represents the backend relational database architecture structured in Third Normal Form (3NF) to ensure absolute referential integrity across all core tables.

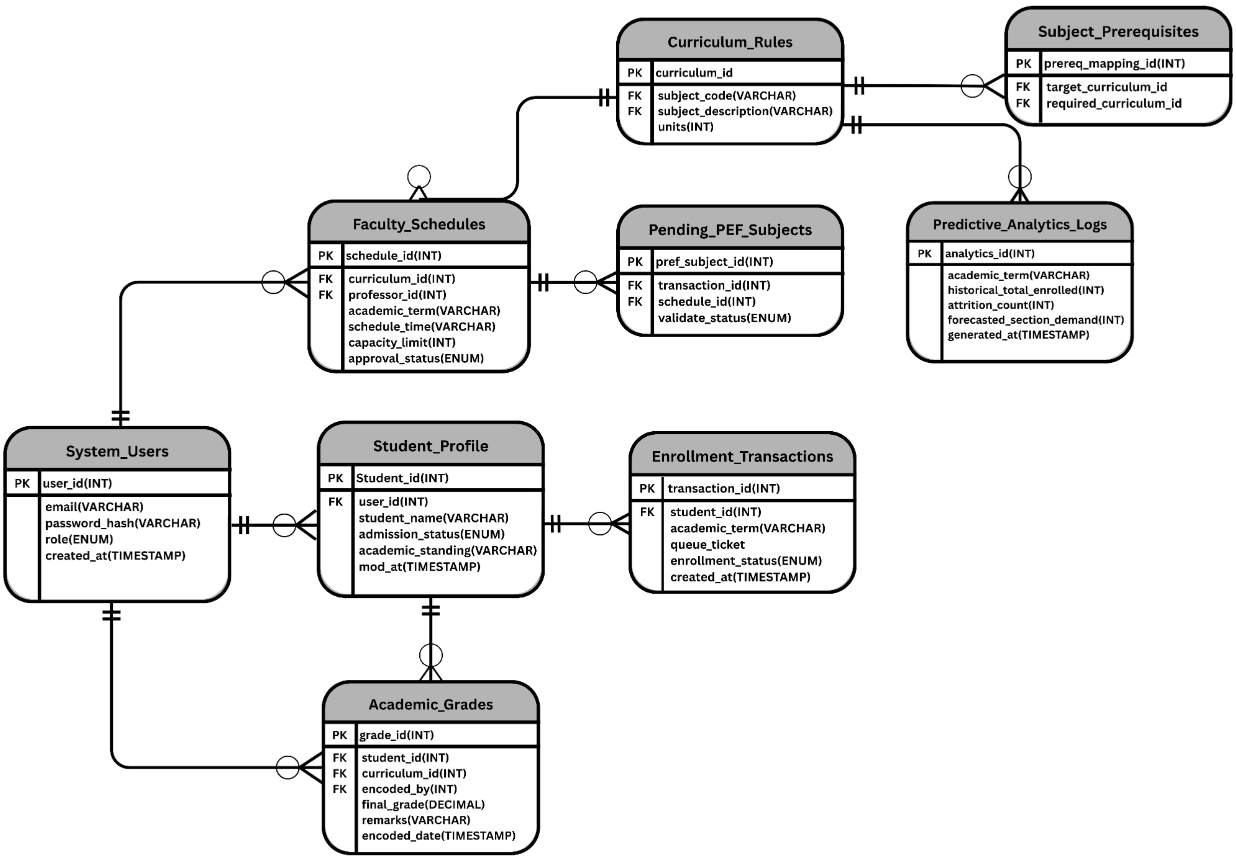

Figure 12. Entity Relationship Diagram of the Proposed System

Figure 12 represents the backend relational database architecture structured in Third Normal Form (3NF) to ensure absolute referential integrity across all core tables. **The schema has been expanded to include dedicated entities for User_Accounts and Compliance_Logs to support secure student portal access and mandatory institutional reporting.** The schema utilizes primary and foreign keys to process academic constraints without data anomalies, specifically isolating historical aggregations within the Predictive_Analytics_Logs entity to ensure that complex forecasting operations do not impact the speed of daily transactional processing. **Furthermore, the normalization of the Academic_Grades and Enrollment_Transactions entities facilitates the automated extraction of Attrition Metrics (Drops, Withdrawals,**

Failures) and Honor Evaluation reports (Dean's List), grounding administrative foresight in a robust and secure data layer.

3.5 Project Evaluation

Evaluation Design

The evaluation of the Automated Enrollment System with Predictive Analytics will utilize a descriptive-evaluation research design to quantitatively assess the system's overall software quality, functional integrity, and predictive accuracy prior to full institutional deployment.

Respondents of the Study

To ensure a rigorous and unbiased assessment, the system will be evaluated by two distinct groups: (1) IT Experts who will rigorously audit the technical architecture, backend relational databases, and predictive algorithms, and (2) the authorized end-users—specifically the Registrar Head, Program Chair, Dean, Admission Staff, and selected students—who will evaluate the system's operational efficiency, role-based access controls, and user interface design.

Evaluation Instrument

The primary evaluation instrument is a structured survey questionnaire strictly adapted from the ISO/IEC 25010 Software Quality Model. This international standard will measure the proposed system against critical software characteristics, specifically Functional Suitability, Performance Efficiency, Usability, Reliability, Security, Maintainability, and Portability.

Furthermore, the predictive analytics forecasting engine (Process 4.0) will be subjected to isolated statistical validation using standard error metrics to guarantee the mathematical precision of its section demand forecasting.

Data Analysis Method and Rating Scale

The empirical data collected from the evaluation instruments will be statistically analyzed using the Weighted Mean to determine the overall system acceptability and performance level. To systematically interpret the gathered quantitative data, the researchers will utilize a structured 5-point Likert Scale with the following descriptive values:

Table 2. 5-Point Likert Rating Scale

Rating Scale	**Numerical Values**	**Descriptive Value**
5	4.21 - 5.00	Highly Acceptable
4	3.41 - 4.20	Acceptable
3	2.61 - 3.40	Neutral
2	1.81 - 2.60	Unacceptable

1	1.00 - 1.80	Totally Unacceptable

Evaluation Criteria

The proposed system will be rigorously evaluated based on the eight (8) core software quality characteristics strictly defined by the ISO/IEC 25010 standard. The evaluation instrument will measure the system against the following criteria to ensure its readiness for institutional deployment:

Table 3. ISO/IEC 25010 Software Evaluation Criteria

Software Criterion	Description / Focus of Evaluation
Functional Suitability	Evaluates if the system successfully automates prerequisite validation and executes predictive analytics without logical errors.
Reliability	Assesses the system's fault tolerance, maturity, and availability, especially during peak enrollment periods.
Performance Efficiency	Measures the system's response time, processing speed, and resource utilization when executing predictive algorithms and database queries.

Usability	Evaluates the intuitiveness, high-fidelity user interface design, and overall ease of navigation of the digital enrollment portal.
Security	Assesses the strict enforcement of the Role-Based Access Control (RBAC) architecture and the protection of the centralized student records.
Compatibility	Evaluates how seamlessly the web-based platform operates across different web browsers and institutional hardware environments.
Maintainability	Measures the modularity and analytical structure of the system's backend architecture for future administrative modifications.
Portability	Assesses the ease of deploying and installing the application into the local or cloud server infrastructure of the institution.

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